- Microsoft provides the tooling built into ASP.NET Core for supplying the metadata about your API endpoints.
- Swashbuckle is an open-source project that uses the ASP.NET Core metadata to generate an OpenAPI-compatible document.
- NSwag is an open-source project that takes an OpenAPI-compatible document and generates clients (and has many other features!).
- Swagger UI is an open-source project for interacting with APIs based on the OpenAPI document.

Some of these projects have direct dependencies on others (everything depends on the OpenAPI specification, for example), but they may evolve at difference paces. If Swashbuckle doesn't support some new feature of the OpenAPI specification, it won't appear in your documents, and NSwag won't be able to use it.

Most of the tools provide ways to override the behavior to work around these rough edges, but the reality is that if you're using newer or less popular features, you may have more difficulty persuading all the tools in your tool chain to play together nicely.

Overall, the important thing to remember is that OpenAPI documents *may* work well if you have simple requirements or want to use Swagger UI only for testing. In these cases, there's little investment required to add OpenAPI support, and it can improve your workflow, so you might find it worthwhile to try.

If you have more complex requirements, are creating an API that OpenAPI can't easily describe or aren't a fan of the code generation, it may not be worth your time to invest heavily in OpenAPI for your documents.

TIP If you're a fan of code generation but prefer more of a remote procedure call (RPC) style of programming, it's worthwhile to look at gRPC. Code generation for gRPC is robust, supported across multiple languages, and has great support in